

ARPIT KUMAR SINGH

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Education

IIT (BHU), Varanasi

B.Tech in Mining Engineering

Graduated May 2025

Varanasi, Uttar Pradesh

Technical Skills

Languages: C, C++, JavaScript, Python, SQL

AI/ML: LangChain, LangGraph, Advanced RAG Architectures, Hybrid Search, Embeddings, OpenAI API

Web & Backend: React.js, Node.js, Express.js, TailwindCSS, Three.js, React-Three, Framer-Motion, REST APIs

Databases: MongoDB, PostgreSQL, QdrantCloud, Cloudinary

DevOps & Tools: Git, GitHub, Docker

IT Constructs: Data Structures and Algorithms, Object Oriented Programming, Operating Systems, DBMS

Experience

Backend Developer

November 2025 – Present

SparkCognition India Private Limited

Onsite

- Developed a **scalable Django Ninja** email **notification service** using **Mailgun** API to automate event-driven emails for process completion, failure, and audit report generation, integrating **GCP** Cloud Storage to automatically upload generated excel reports to buckets enabling secure, persistent storage and efficient retrieval of reports.
- Designed and maintained **RESTful APIs** across multiple modules, following clean architecture principles and **SOLID** design patterns. Optimized performance using **batch processing**, **in-memory** lookup maps, and **selective field retrieval** to reduce memory overhead and improve response time. Ensured data consistency with atomic transactions.
- Optimized Excel report generation for **100,000+** records using **batched processing** and **in-memory lookups**, cutting DB calls and reducing **API Latency** by **50%**, added **hidden**, **locked ID columns** for reliable row-level tracking.

Backend Developer

June 2025 – October 2025

Zetexa Global Private Limited

Onsite

- Developed an AI-powered chatbot backend using **GPT-4.1-mini**, **text-embedding-small** for embeddings, and QdrantCloud, optimizing retrieval latency and ensuring fast, accurate responses across large internal datasets.
- Built workflow orchestration using **LangGraph** for query routing, handling general information and triggering custom functions for dynamic purchase URL generation, with **Zod** schema validation for type-safe request and response.
- Integrated the chatbot to **Zoho SalesIQ** using **Deluge** scripting for live chat, ticketing, and real-time assistance. Designed and developed secure **REST APIs** to allow employees to upload, manage and update new documents.

Projects

3D Photo Album Web App | *React Three/Fiber, React Router, React Dropzone, OpenGLSL*

April 2025

- Developed a full-stack web application with an interactive 3D photo experience to store and view photos on the web app using **Three.js**, **React-Three/Fiber**, **React-Three/drei**, and **OpenGLSL**, rendering a 3D album layout in WebGL
- Built a photo upload system using React Dropzone for drag-and-drop functionality and **Cloudinary**, added animations using **Framer Motion**, and integrated **Auth0** for secure authentication and route protection. Implemented navigation with **React Router-DOM** and designed a responsive UI using **TailwindCSS** for a better user experience.

LegalMind AI | *React.js, Framer-Motion, Langchain, TailwindCSS, Qdrant Cloud*

May 2026

- Designed and implemented a **hybrid Retrieval-Augmented Generation (RAG)** pipeline combining **BM25** keyword search and dense semantic embeddings (**Sentence-Transformers**) with cross-encoder re-ranking.
- **Orchestrated** a stateful, multi-node retrieval workflow using **LangGraph**, executing parallel **sparse (BM25)** and **dense vector search** against a **Qdrant** vector database, followed by **cross-encoder-based re-ranking** of documents and **LLM-driven response synthesis** to deliver contextually accurate results with reduced hallucination.
- Engineered a highly responsive frontend interface using **ReactJS** for component-based architecture, **Tailwind CSS** for utility-first styling, and **Framer Motion** for smooth animations and transitions, enhancing overall user engagement.

AI-powered 3D Portfolio | *React.js, React-Three, Framer-Motion, Three.js, Node.js, LangChain* January 2026

- Built a personalized portfolio website using **React.js**, **React Three Fiber**, and **Three.js**, incorporating interactive 3D models seamlessly integrated into the site. Optimized rendering performance using Lazy loading and Suspense.
- Integrated an AI assistant using **LangChain** and **Node.js**, powered by Llama 3.3-70b for intelligent interactions.
- Implemented **Framer Motion** for smooth animations and styled the UI using **TailwindCSS** for a responsive design.